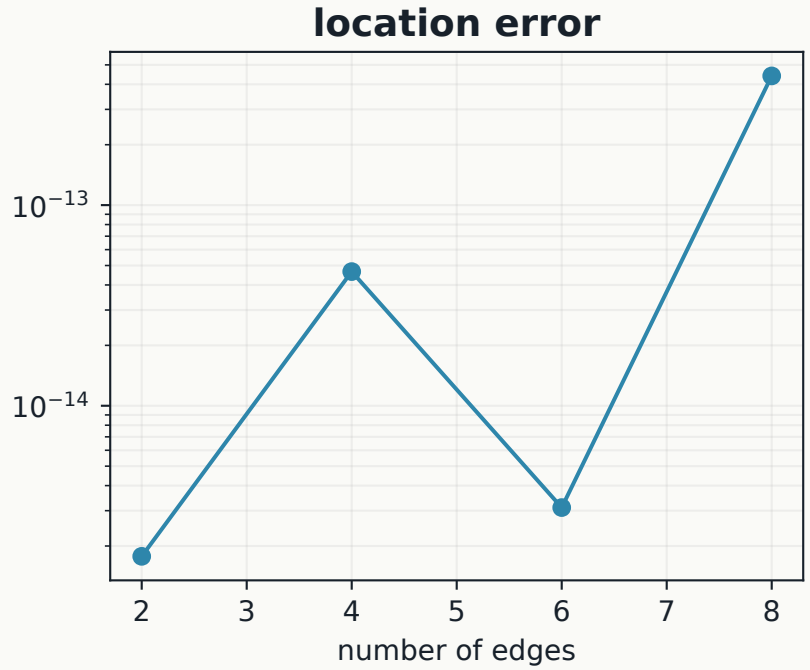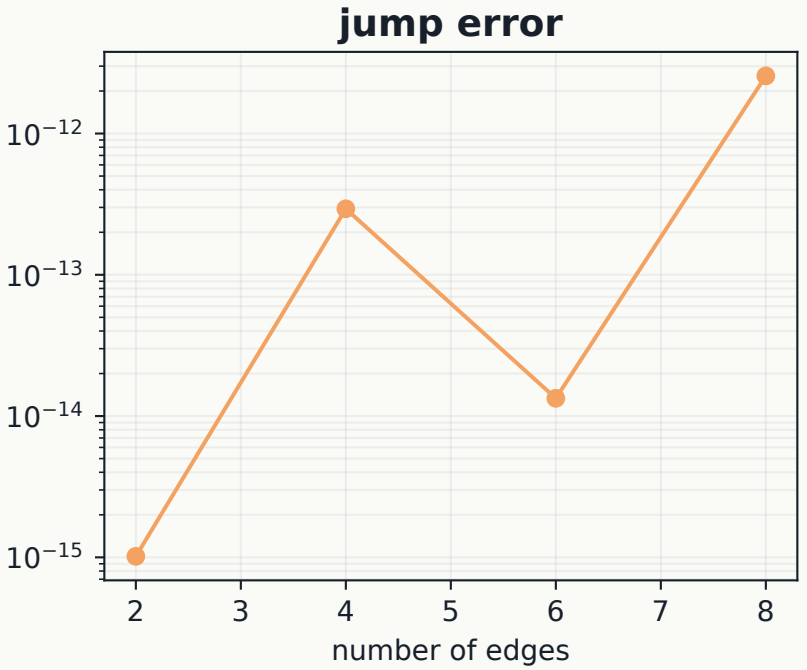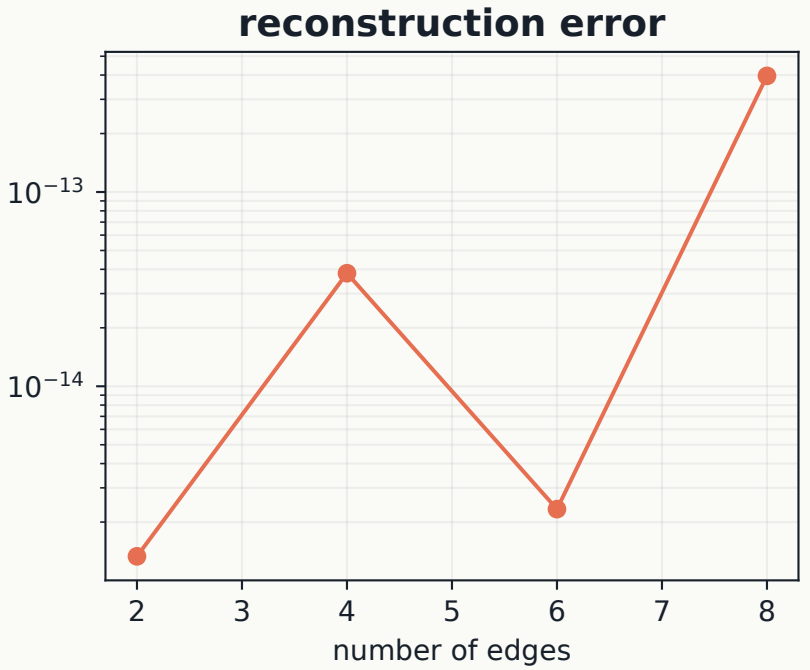Exact off-grid recovery across several unknown-edge counts